

Magnetic Phases and How Machine Learning Learns Them

A working reference for a physics + AI collaboration

This document has two goals. First, to give you a physicist’s map of the magnetic phases – what they are, what distinguishes them, and the models that produce them. Second, to show precisely where and how machine learning attaches to each problem, so that the division of labour in a physics/ML collaboration is concrete rather than vague. Physics is treated rigorously; the ML is treated pedagogically. Wherever a figure could be computed rather than drawn by hand, it was.

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1 The starting point: spins, exchange, and models

A magnet is a lattice of local magnetic moments (“spins”) $\mathbf{S}_i$ that interact. Almost all of equilibrium magnetism follows from one deceptively simple energy, the *exchange Hamiltonian*,

$$H = - \sum_{\langle ij \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j - \sum_i \mathbf{B} \cdot \mathbf{S}_i, \quad (1)$$

where $\langle ij \rangle$ runs over interacting pairs (usually nearest neighbours), J_{ij} is the exchange coupling, and $\mathbf{B}$ an external field. The sign and structure of J_{ij} decides almost everything:

- $J > 0$ favours *parallel* neighbours $\Rightarrow$ ferromagnetic tendency;
- $J < 0$ favours *antiparallel* neighbours $\Rightarrow$ antiferromagnetic tendency;
- random-sign $J_{ij} \Rightarrow$ frustration and glassiness;
- an antisymmetric relativistic correction, the Dzyaloshinskii–Moriya term $\mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j)$, favours *twisting* $\Rightarrow$ skyrmions.

The *number of components* of $\mathbf{S}_i$ defines the standard models, and this choice controls which phases and transitions are even possible:

Model	Spin	Symmetry	Signature physics
Ising	$S_i = \pm 1$ (1 comp.)	discrete $\mathbb{Z}_2$	sharp order, 2D exact (Onsager)
XY	angle θ_i (2 comp.)	continuous $O(2)$	vortices, BKT transition
Heisenberg	unit vector (3 comp.)	continuous $O(3)$	spin waves, no order in 2D
Kitaev	bond-dependent Ising	anisotropic	exact quantum spin liquid

In the quantum case the $\mathbf{S}_i$ are operators and the ground state lives in a Hilbert space of dimension 2^N – the exponential wall that later forces us toward neural-network representations.

2 Order, broken symmetry, and the order parameter

A *phase* is a region of parameter space (temperature T , field B , coupling ratios) in which the system organises the same way. Phases are separated by *phase transitions*, and most magnetic phases are distinguished by an *order parameter*: a quantity that is zero in the disordered phase and nonzero once a symmetry is spontaneously broken. For a ferromagnet it is the magnetization

$$m = \frac{1}{N} \left\langle \sum_i S_i \right\rangle, \quad (2)$$

for an antiferromagnet it is the *staggered* magnetization $m_s = \frac{1}{N} \langle \sum_i (-1)^i S_i \rangle$, and for exotic phases it can be far more subtle (a chirality, a topological winding number, or – in a spin liquid – *no local order parameter at all*). The central practical difficulty, and the reason ML is useful, is exactly this: **we do not always know what the order parameter is**, and hand-crafting one for a new exotic phase can take years.

3 A gallery of magnetic phases

Figure 1 shows the four “collinear” phases as spin snapshots; the noncollinear and topological phases follow.

Paramagnet (disordered). At high T thermal fluctuations win; spins point every which way and average to zero, $m = 0$. No symmetry is broken. Susceptibility follows the Curie–Weiss law $\chi \propto 1/(T - \theta_{CW})$.

Ferromagnet. Below the Curie temperature T_c the spins align, giving $m \neq 0$ and macroscopic magnetization. The $\mathbb{Z}_2$ (up/down) or $O(n)$ rotational symmetry of Eq. (1) is spontaneously broken. Examples: Fe, Co, Ni.

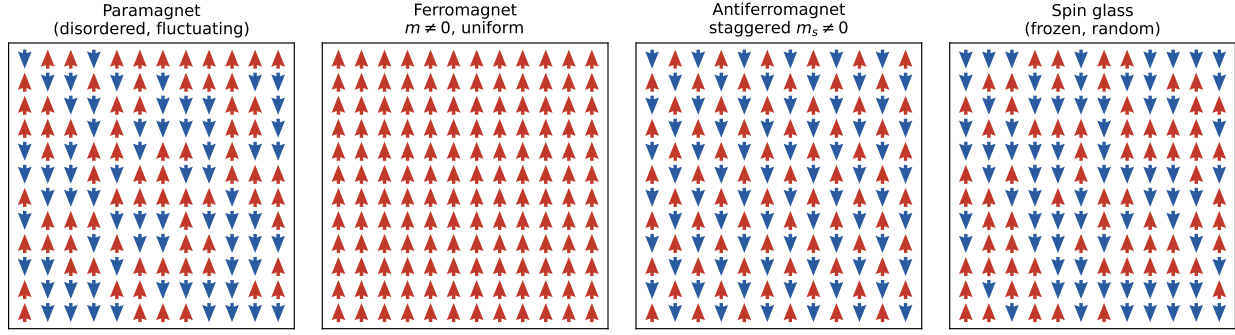


Figure 1: The four collinear phases as Ising spin configurations (red = up, blue = down). The paramagnet and the spin glass look alike in a single snapshot – their difference is *dynamical*: paramagnet spins fluctuate freely, spin-glass spins are frozen into one random pattern. This is precisely the kind of distinction a learned representation can capture and a naive eye cannot.

Antiferromagnet and ferrimagnet. With $J < 0$, neighbours anti-align into a staggered (Néel) pattern; the net moment is zero but $m_s \neq 0$ below the Néel temperature T_N . If the two sublattices have unequal moments the leftover is a *ferrimagnet* (e.g. magnetite). Examples: MnO, NiO.

Frustration and noncollinear order. When the lattice geometry forbids satisfying every bond simultaneously the system is *frustrated* (Fig. 2a). The classic case is three antiferromagnetically coupled spins on a triangle: no arrangement makes all three pairs antiparallel. The compromise is often a noncollinear state such as the 120° order of the triangular-lattice antiferromagnet (Fig. 2b). Frustration is the gateway to spin glasses, spin liquids and skyrmions.

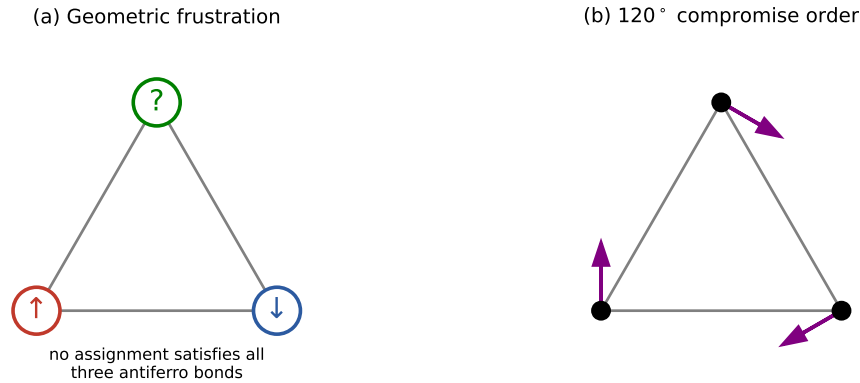


Figure 2: Geometric frustration. (a) On a triangle with antiferromagnetic bonds the third spin cannot satisfy both of its neighbours. (b) The system resolves the conflict with a noncollinear 120° compromise – an ordered but non-trivial ground state.

Spin glass. With *quenched randomness* in the couplings (random J_{ij} of both signs, as in the Edwards–Anderson model) the spins freeze into a random, frustrated pattern with no periodic order. The order parameter is not a magnetization but the Edwards–Anderson overlap $q_{EA} = \frac{1}{N} \sum_i \langle S_i \rangle^2$, measuring how frozen each spin is. Spin glasses have a rugged free-energy landscape with many

nearly degenerate minima – the reason they are computationally brutal (and, not coincidentally, why spin-glass theory underlies both optimization and neural networks).

Skyrmions: topological textures. The Dzyaloshinskii–Moriya interaction twists neighbouring spins, stabilising nanoscale swirls called *magnetic skyrmions* (Fig. 3). A skyrmion is *topologically protected*: its spins wrap the unit sphere an integer number of times,

$$Q = \frac{1}{4\pi} \int \mathbf{m} \cdot (\partial_x \mathbf{m} \times \partial_y \mathbf{m}) dx dy \in \mathbb{Z}, \quad (3)$$

and cannot be smoothly unwound. In chiral magnets (MnSi, FeGe) skyrmions crystallise into a lattice inside a small pocket of the field–temperature phase diagram (Fig. 3). Because they are small, stable and current-drivable, they are a leading candidate for racetrack memory – the applied, device-physics end of magnetism.

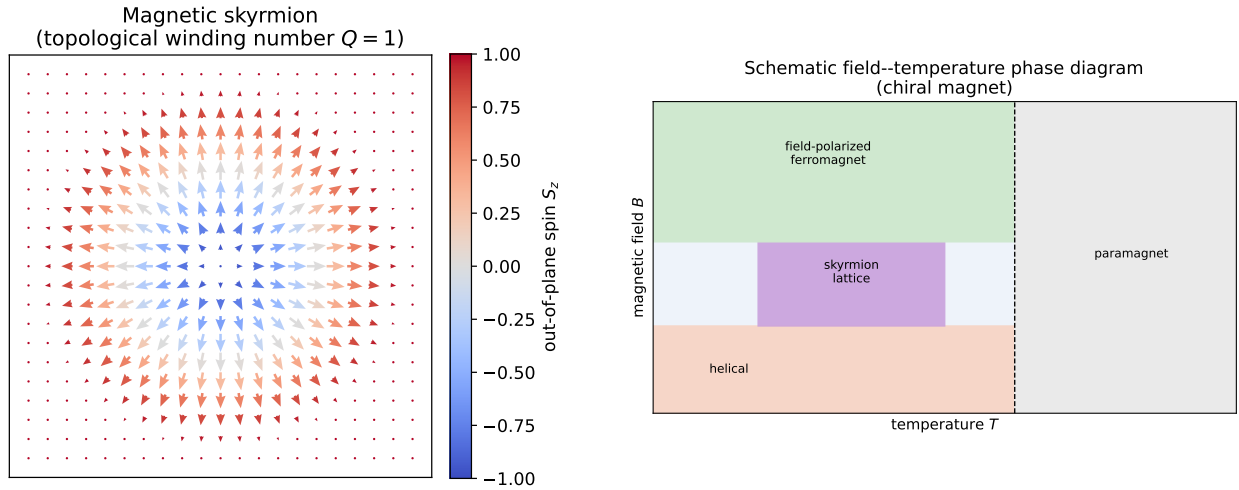


Figure 3: *Left:* a Néel skyrmion – in-plane spins (arrows) rotate from down at the core to up at the rim; colour is the out-of-plane component S_z . *Right:* skyrmions occupy a small pocket of the B – T phase diagram of a chiral magnet, bordered by helical/spiral, field-polarised, and paramagnetic phases.

Quantum spin liquid. In strongly frustrated *quantum* magnets, quantum fluctuations can prevent order down to $T = 0$: the spins remain collectively entangled and fluctuating – a *spin liquid*. There is no local order parameter; the phase is diagnosed by long-range entanglement, fractionalised excitations, and topological order. Candidate materials include the kagome antiferromagnet herbertsmithite and Kitaev material α - RuCl_3 . This is the hardest phase to identify, classically or with ML, and a major frontier.

4 Phase transitions and criticality

Approaching T_c from below, the order parameter vanishes continuously and response functions diverge (Fig. 4). Near a continuous transition, observables obey *power laws* governed by *critical exponents*,

$$m \sim (T_c - T)^\beta, \quad \chi \sim |T - T_c|^{-\gamma}, \quad \xi \sim |T - T_c|^{-\nu}, \quad (4)$$

where ξ is the correlation length. The susceptibility is fixed by fluctuations, $\chi = \beta(\langle M^2 \rangle - \langle M \rangle^2) / N$. Remarkably, the exponents depend only on symmetry and dimension, not microscopic detail – *universality*. The 2D Ising model is exactly solvable (Onsager 1944) with $\beta = 1/8$, $\gamma = 7/4$, $\nu = 1$, which is why it is the perfect ML benchmark: the ground truth is known in closed form.

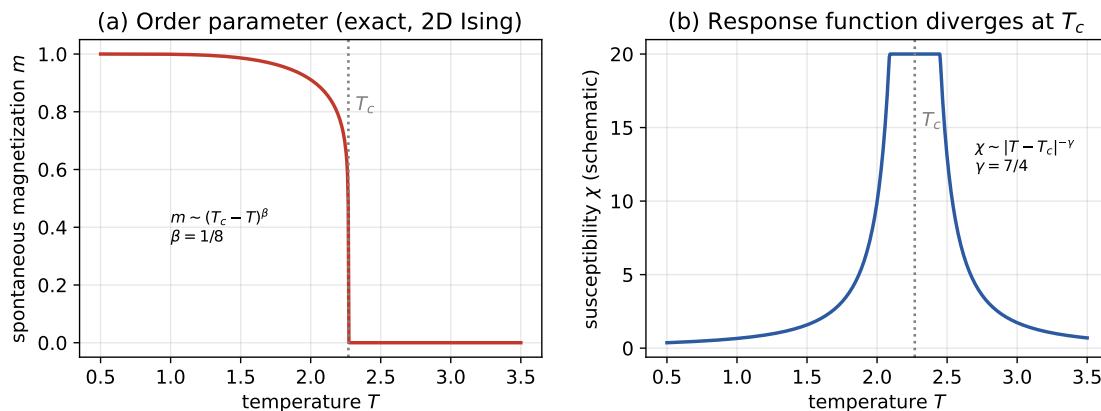


Figure 4: (a) Exact 2D-Ising spontaneous magnetization $m(T) = [1 - \sinh^{-4}(2\beta J)]^{1/8}$ vanishing at T_c with exponent $\beta = 1/8$. (b) The susceptibility diverges at T_c (schematic $\gamma = 7/4$).

A transition without a local order parameter: BKT. The 2D XY model has a continuous symmetry and *no* conventional ordered phase, yet it undergoes the Berezinskii–Kosterlitz–Thouless transition driven by the binding and unbinding of *vortices* (Fig. 5). Below T_{BKT} vortices bind into neutral pairs; above it they proliferate and destroy quasi-order. BKT is the paradigm of a *topological* phase transition – and precisely the kind of subtle transition where an unsupervised network, having no built-in notion of “magnetization”, can still detect the change.

XY model: a vortex--antivortex pair (BKT physics)

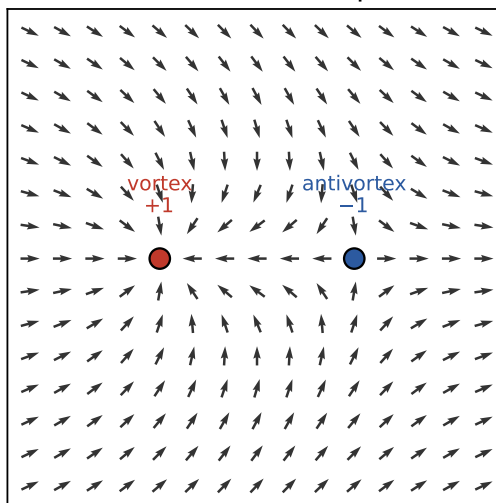


Figure 5: A vortex (+1) and antivortex (−1) in the XY model. The BKT transition is the unbinding of such pairs; there is no local order parameter, so it defies naive order-parameter thinking and motivates learned detectors.

Critical slowing down – the crux for computation. Near T_c the correlation length ξ diverges, and any *local* Monte Carlo update (flip one spin, accept/reject) takes a diverging number of steps to produce an independent configuration:

$$\tau_{\text{int}} \sim \xi^z \sim |T - T_c|^{-\nu z}, \quad (5)$$

with dynamic exponent $z \approx 2$ for single-spin dynamics. Independent samples become extraordinarily expensive exactly where the interesting physics lives. This single fact is the doorway through which one class of ML methods enters (Sec. 6.3).

5 Why classical tools struggle (and where ML helps)

Before ML, three workhorses dominate, each with a hard wall:

- **Monte Carlo** samples $p(s) \propto e^{-\beta E(s)}$ with a Markov chain. Wall: *critical slowing down* near T_c (partly cured by cluster algorithms such as Wolff/Swendsen–Wang, but not for generic frustrated or continuous-spin models).
- **Exact diagonalization** of quantum models is exact but limited to ~ 40 spins by the 2^N Hilbert space.
- **Quantum Monte Carlo** handles large quantum systems – until the *sign problem* makes frustrated or fermionic models exponentially hard.

Machine learning does not repeal these walls, but it offers new *representations* and *samplers* that sometimes scale better or need no hand-crafted order parameter. The rest of the document maps those methods.

6 How AI attaches to magnetism: a taxonomy

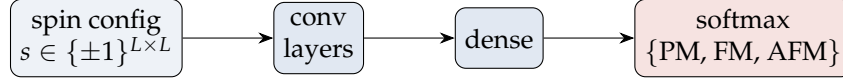
There are five distinct ways ML enters, and they answer genuinely different questions. Table 1 is the summary; the subsections give the mechanics.

Table 1: Five ways machine learning attaches to magnetic models.

Method	Question it answers	How	Phase target
Supervised classifier (CNN/GNN)	“Which phase is this configuration?”	Train on labelled snapshots; net learns the decision boundary	FM/AFM/PM
Unsupervised (PCA, VAE)	“Is there a transition, and what is the order parameter?”	Learn latent structure without labels; clusters = phases	any, incl. hidden
Generative sampler (VAN)	“Give me independent samples and the free energy.”	Autoregressive net trained to variational free energy	classical, near T_c
Neural quantum state	“What is the ground state of this quantum magnet?”	Wavefunction = neural net, minimise energy	spin liquids
PINN / neural operator	“Solve the magnetisation dynamics / screen materials.”	Encode LLG PDE as a loss; or map structure→property	skyrmions, discovery

6.1 Supervised phase classification

The most direct idea (Carrasquilla & Melko 2017): treat a spin configuration as an image, label it by its known phase, and train a convolutional network to predict the label. The network discovers the relevant features on its own, and the point where its output probability crosses 1/2 pinpoints T_c .



Strength: simple, accurate. *Weakness:* needs labels, i.e. you must already know the phases – so it confirms rather than discovers.

6.2 Unsupervised discovery of order parameters

Here we hand the network *unlabelled* configurations and ask it to find structure. The simplest such method, principal component analysis, already works for the Ising model: the leading principal component of raw spin snapshots *is* the magnetization (Wang 2016). Figure 6 shows this computed from scratch – PCA was never told about temperature or magnetization, yet PC1 recovers the order parameter and separates the phases.

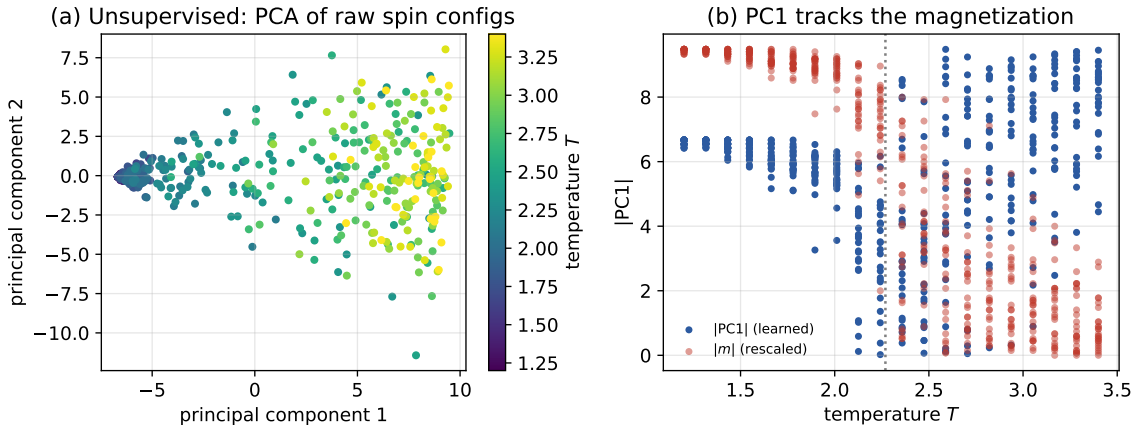


Figure 6: Unsupervised learning, computed here on Monte Carlo Ising configurations ($L = 16$). (a) Projecting raw spin snapshots onto their first two principal components; colour is temperature, which the algorithm never saw. Ordered (cold) configurations spread along PC1; disordered (hot) ones collapse near the origin. (b) $|PC1|$ tracks the true magnetization and drops near T_c – a *machine-discovered* order parameter.

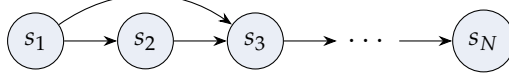
Nonlinear versions – autoencoders and *variational autoencoders* (VAEs) – extend this to phases where PCA fails, learning a low-dimensional latent code whose clustering reveals the phase diagram. This is the ideal “discovery” tool for exotic order (chiral, topological) where no human order parameter is known: the physicist supplies the Hamiltonian and the ground truth; the network proposes the coordinate.

6.3 Generative sampling: Variational Autoregressive Networks

This is the method that directly attacks critical slowing down, and the core of our joint project. Instead of a Markov chain we build a normalised neural distribution $q_\theta(s)$ written *autoregressively*,

so that each spin's probability is conditioned on the previous ones:

$$q_\theta(s) = \prod_{k=1}^N q_\theta(s_k \mid s_1, \dots, s_{k-1}). \quad (6)$$



each conditional = a network output

Two properties fall out for free: we can **sample exactly** (one sequential pass, i.i.d. configurations) and **evaluate** $q_\theta(s)$ **exactly**. We train by minimising the *variational free energy*

$$\beta F_q = \mathbb{E}_{s \sim q}[\beta E(s) + \ln q_\theta(s)] = \beta F_{\text{exact}} + D_{\text{KL}}(q \parallel p) \geq \beta F_{\text{exact}}, \quad (7)$$

which is a *rigorous upper bound* – the network hands you a certified free energy, something ordinary MC does not. Because spins are discrete we use the score-function (REINFORCE) gradient with a baseline b ,

$$\nabla_\theta(\beta F_q) = \mathbb{E}_{s \sim q}[(\beta E(s) + \ln q_\theta(s) - b) \nabla_\theta \ln q_\theta(s)]. \quad (8)$$

Since every sample is independent, the autocorrelation time is $\tau_{\text{int}} = 1$ at *all* temperatures – critical slowing down is gone by construction (Fig. 7). Figure 8 shows our own $L = 8$ run reproducing the exact Onsager thermodynamics.

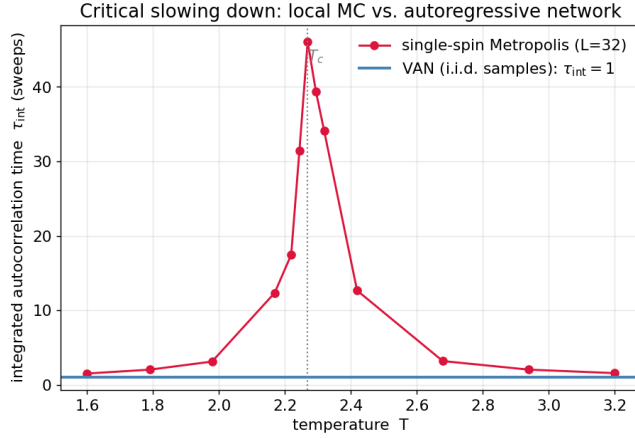


Figure 7: Critical slowing down made visible (our computation). Single-spin Metropolis autocorrelation time τ_{int} spikes at T_c (here to ~ 46 sweeps, $L = 32$); a variational autoregressive network draws independent samples, so $\tau_{\text{int}} = 1$ everywhere.

The bridge to radiation transport. VAN training is *importance sampling with a learned proposal*: q_θ is the biased sampling distribution and $\ln q$ is the log-weight – the same weight bookkeeping as weight-window variance reduction in Monte Carlo particle transport, but in configuration space instead of phase space. The variational free-energy bound is the analogue of an unbiased weighted estimator. This is a genuine conceptual overlap with deep-penetration shielding work, not a superficial one.

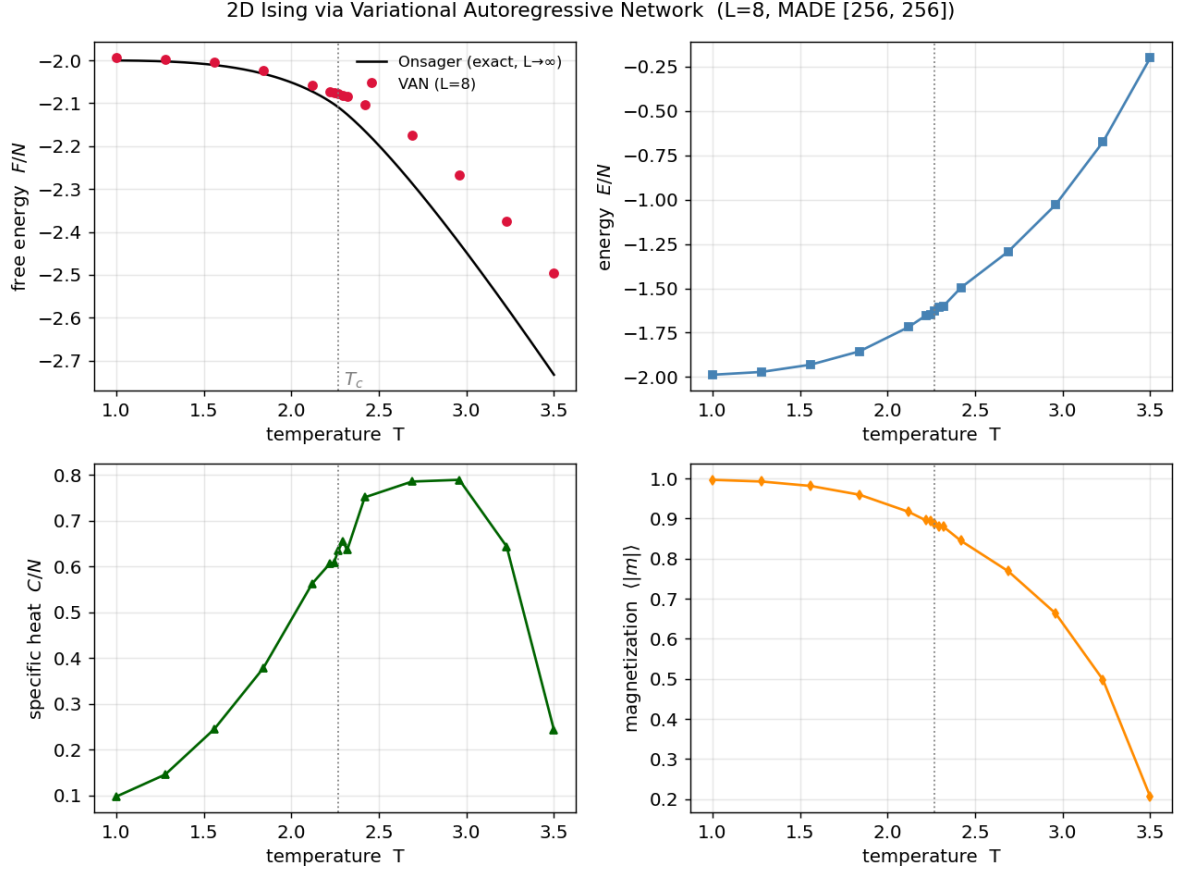


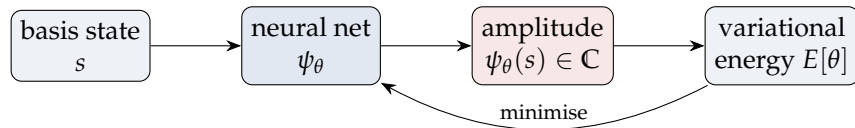
Figure 8: Our from-scratch VAN ($L = 8$) versus the exact Onsager solution: the free energy respects the variational bound (points sit above the exact curve) and the energy, specific heat and magnetization reproduce the ordering crossover near T_c .

6.4 Neural quantum states: ground states of quantum magnets

For *quantum* magnets the object of interest is the ground-state wavefunction $\psi(s) = \langle s | \psi \rangle$ over the 2^N basis states. Carleo & Troyer (2017) proposed representing $\psi_\theta(s)$ by a neural network and minimising the variational energy

$$E[\theta] = \frac{\langle \psi_\theta | H | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle} \geq E_0, \quad (9)$$

by stochastic sampling of $|\psi_\theta(s)|^2$.



The frontier target is the frustrated / spin-liquid regime where the wavefunction has a hard *sign structure* (the “sign problem” resurfacing). The ML challenges are architectures that enforce lattice and spin symmetries and represent signs – restricted Boltzmann machines, autoregressive/RNN wavefunctions, and transformers. This is the highest-ceiling, most competitive direction; enter it only with a physics insight others lack.

6.5 PINNs, neural operators, and materials discovery

Two applied directions round out the map. **Physics-informed neural networks** (Raissi et al. 2019) solve the Landau–Lifshitz–Gilbert equation of micromagnetics,

$$\frac{d\mathbf{m}}{dt} = -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt}, \quad (10)$$

by encoding the PDE residual as a training loss – useful for skyrmion nucleation and domain dynamics; *neural operators* instead learn the whole map from material parameters to the equilibrium texture, replacing expensive solvers. **Graph neural networks** (Xie & Grossman 2018) take a crystal structure as a graph and predict magnetic ground state and ordering temperature, enabling high-throughput screening for rare-earth-free room-temperature ferromagnets; large-scale efforts of this kind (e.g. GNoME, 2023) have expanded known stable materials by orders of magnitude.

7 Putting it together: our roadmap

The methods above are not equally ripe for a two-person team. Ordered by realism:

1. **Paper 1 – VAN for the classical Ising model, scaled up** ($L = 16, 32, 64$). Headline: elimination of critical slowing down plus a certified free-energy bound. Working code and the figures in this document already exist.
2. **Paper 2 – the XY model and the BKT transition**. Same machinery, a continuous-spin energy, a topological transition with no local order parameter – higher physics interest, modest new ML.
3. **Stretch – neural quantum states for a frustrated magnet**. High ceiling, crowded field; pursue only where a specific physics insight gives an edge.

Division of labour. *Physics (you)*: define Hamiltonians and phase diagrams, guarantee ground truth (Onsager, enumeration, exact diagonalization), interpret what the network learns, and own the importance-sampling / variance-reduction narrative. *ML (collaborator)*: architectures (MADE $\rightarrow$ PixelCNN $\rightarrow$ transformer; symmetry-equivariant layers), optimisation (REINFORCE variance reduction, natural gradient, schedules), and scaling/engineering.

Appendix: one-line glossary

- **Order parameter** – zero in the disordered phase, nonzero once a symmetry breaks.
- **Universality** – critical exponents depend only on symmetry and dimension.
- **Frustration** – geometry/couplings forbid satisfying all bonds at once.
- **Critical slowing down** – local MC decorrelation time diverges at T_c ($\tau \sim \xi^z$).
- **Sign problem** – oscillating signs make frustrated/fermionic QMC exponentially hard.
- **Variational free energy** – $\beta F_q = \mathbb{E}_q[\beta E + \ln q] \geq \beta F_{\text{exact}}$; the VAN loss.
- **Topological order / winding** – integer-valued, robust to smooth deformation (skyrmion Q , vortex charge).

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